

Homework #4

Math 104: Intro to Analysis

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0 Various Theorems

0.1 Limit Proof I

For $p, q, \alpha, C \in \mathbb{R}$ where $\alpha > 1$ and $C > 0$, $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$.

Proof. Let $\varepsilon > 0$. Let $\delta = \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$, which is greater than zero because $\alpha - 1 > 0$ and $C > 0$. Suppose that $0 < |p - q| < \delta$. Substitute in our value of δ to find $0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}}$. Now, raise all sides to the positive power $\alpha - 1$, then multiply by C .

$$0 < |p - q| < \alpha^{-1} \sqrt[\alpha]{\frac{\varepsilon}{C}} \quad (1)$$

$$0 < |p - q|^{\alpha-1} < \frac{\varepsilon}{C} \quad (2)$$

$$0 < C |p - q|^{\alpha-1} < \varepsilon \quad (3)$$

Since $0 < C |p - q|^{\alpha-1}$, $C |p - q|^{\alpha-1} = \left| C |p - q|^{\alpha-1} \right|$. Therefore, $\left| C |p - q|^{\alpha-1} \right| < \varepsilon$. Ergo, $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$. OEΔ

0.2 Limit Proof II

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad (4)$$

Proof. Let $\varepsilon > 0$. Let $\delta = \frac{1}{\varepsilon} > 0$. Suppose that $x > \delta$. Multiply both sides by the reciprocals of x and δ to find that $\frac{1}{\delta} > \frac{1}{x}$. Substitute in the value of δ to find that $\frac{1}{\frac{1}{\varepsilon}} = \varepsilon > \frac{1}{x}$. Since $\delta > 0$ and $x > \delta$, $x > 0$, meaning $\frac{1}{x} > 0$, so $\frac{1}{x} = \left| \frac{1}{x} \right|$. So $\varepsilon > \left| \frac{1}{x} \right|$. Therefore, $x > \delta$ implies $\varepsilon > \left| \frac{1}{x} \right|$. Ergo, $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. OEΔ

0.3 Alternate Derivative Definition

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable.

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (5)$$

Proof. Recall the definition of $\frac{df(x)}{dx}$.

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} \quad (6)$$

This means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that $|c - x| < \delta$ implies $|f(c) - f(x)| < \varepsilon$. Let $h = c - x$. Solving for c , $c = x + h$. We can therefore conclude that $|h| < \delta$ implies $|f(x+h) - f(x)| < \varepsilon$. This fits a definition of a new limit.

$$\frac{df(x)}{dx} = \lim_{c \rightarrow x} \frac{f(c) - f(x)}{c - x} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (7)$$

This completes the proof. OEΔ

0.4 Alternate Mean Value Theorem

Let f be a real, continuous function on $[x, x+h]$ which is differentiable in $(x, x+h)$, then there is a point $b \in (x, x+h)$ at which

$$f(x+h) - f(x) = hf'(b). \quad (8)$$

Proof. Recall the Mean Value Theorem. If f is a real continuous function on $[x, y]$ which is differentiable in (x, y) , then there is a point $b \in (x, y)$ at which

$$f(y) - f(x) = (y-x)f'(b) \quad (9)$$

Let $h = y - x$. By extension, $y = x + h$. Therefore, for $b \in (x, x+h)$,

$$f(x+h) - f(x) = hf'(b) \quad (10)$$

OEΔ

Let f be a real, continuous function on $[x-h, x]$ which is differentiable in $(x-h, x)$, then there is a point $c \in (x-h, x)$ at which

$$f(x) - f(x-h) = hf'(c). \quad (11)$$

Proof. This is very similar to the last. Recall the Mean Value Theorem. If f is a real continuous function on $[y, x]$ which is differentiable in (y, x) , then there is a point $c \in (y, x)$ at which

$$f(x) - f(y) = (x-y)f'(c) \quad (12)$$

Let $h = x - y$. By extension, $y = x - h$. Therefore, for $c \in (x-h, x)$,

$$f(x) - f(x-h) = hf'(c) \quad (13)$$

OEΔ

Note that these are respectively equivalent to:

$$f'(b) = \frac{f(x+h) - f(x)}{h} \quad (14)$$

$$f'(c) = \frac{f(x) - f(x-h)}{h} \quad (15)$$

0.5 Mean in the Middle

For $a, b \in \mathbb{R}$, if $a < b$, then $a < \frac{a+b}{2} < b$.

Proof. Let $a, b \in \mathbb{R}$ where $a < b$. Add b to both sides and find $a+b < 2b$. Divide both sides by two to find $\frac{a+b}{2} < b$.

Take $a < b$. Add a to both sides and find $2a < a+b$. Divide both sides by two to find $a < \frac{a+b}{2}$.

Ergo, by the transistive property, $a < \frac{a+b}{2} < b$. OEΔ

0.6 Lambda in the Middle

Let $0 < \lambda < 1$ and $x, y \in \mathbb{R}$ where $x < y$. $x < \lambda x + (1-\lambda)y < y$.

Proof. Since $y > x$, $(1-\lambda)y > (1-\lambda)x$. Add λx to both sides and find $\lambda x + (1-\lambda)y > \lambda x + (1-\lambda)x = x$.

Since $y > x$, $\lambda y > \lambda x$. Add $(1-\lambda)y$ to both sides and find $y = \lambda y + (1-\lambda)y > \lambda x + (1-\lambda)y$

Transistively, $x < \lambda x + (1-\lambda)y < y$. OEΔ

1 Problem 1

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : (X, d_X) \rightarrow (Y, d_Y)$ is called *Lipschitz continuous* if there is a constant $L > 0$ so that

$$d_Y(f(p), f(q)) \leq L d_X(p, q)$$

for all $p, q \in X$. For $\alpha > 0$, we say f is *Hölder continuous (with exponent α)* if there is a constant $C > 0$ so that

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$$

for all $p, q \in X$. In particular, Hölder continuous with exponent 1 means the same thing as Lipschitz.

- (a) Fix a point $p \in X$. Show that the map $g : X \rightarrow \mathbb{R}$ defined by $g(x) = d(x, p)$ is Lipschitz continuous, with constant $L = 1$.
- (b) Prove that any Hölder continuous function (with any exponent $\alpha > 0$) is uniformly continuous.
- (c) Prove that if f is Lipschitz continuous and bounded, then f is Hölder continuous with exponent α , for any $0 < \alpha \leq 1$.
- (d) Let $[a, b]$ be a closed interval, and suppose $f : [a, b] \rightarrow \mathbb{R}$ is C^1 (meaning its first derivative exists and is continuous on $[a, b]$). Show that f is Lipschitz continuous.
- (e) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is Hölder continuous with exponent $\alpha > 1$. Show that f is constant.

1.1 Solution (a)

Proof. Since the codomain of g is $\mathbb{R}$, we can use the standard metric as d_Y . From the triangle inequality, we know that $d_X(x, y) + d_X(p, x) \geq d_X(p, y)$ and $d_X(x, y) + d_X(p, y) \geq d_X(p, x)$. We can reorder both of these to find that $d_X(x, y) \geq d_X(p, y) - d_X(p, x)$ and $d_X(x, y) \geq d_X(p, x) - d_X(p, y)$. Consolidating these, we can apply the absolute value to find that $d_X(x, y) \geq |d_X(p, y) - d_X(p, x)|$. This is also known as the inverse triangle inequality. Since the absolute value is the usual metric over $\mathbb{R}$, this means that $d_X(x, y) \geq d_{\mathbb{R}}(d_X(p, y), d_X(p, x))$. Since $g(x) = d_X(p, x)$, $d_X(x, y) \geq d_{\mathbb{R}}(g(y), g(x))$. Therefore, g is Lipschitz continuous with constant $L = 1$. OEΔ

1.2 Solution (b)

Proof. Let f be Hölder continuous with exponent α for $\alpha > 0$.

$$d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha \tag{16}$$

Let $\varepsilon > 0$ and let $\delta = \sqrt[\alpha]{\frac{\varepsilon}{C}}$, which allows that $\delta > 0$ since $C > 0$. Suppose that for some $p, q \in X$, $d_X(p, q) < \delta$. We can substitute in our value of δ and isolate ε .

$$d_X(p, q) < \delta = \sqrt[\alpha]{\frac{\varepsilon}{C}} \tag{17}$$

$$d_X(p, q)^\alpha < \frac{\varepsilon}{C} \tag{18}$$

$$C d_X(p, q)^\alpha < \varepsilon \tag{19}$$

Since $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$, this means $d_Y(f(p), f(q)) < \varepsilon$. Therefore, $d_X(p, q) < \delta$ implies $d_Y(f(p), f(q)) < \varepsilon$. Ergo, f is uniformly continuous. OEΔ

1.3 Solution (c)

Proof. Let $0 < \alpha \leq 1$, and let f be Lipschitz continuous. Let $p, q \in X$.

$$d_Y(f(p), f(q)) \leq L d_X(p, q) \quad (20)$$

Since f is bounded, then there exists a number $M \in \mathbb{R}$ and a point $f(q) \in Y$ for some $q \in X$ such that for all $p \in X$, $d_Y(f(p), f(q)) < M$. Suppose we take three cases: $d_X(p, q) = 1$, $d_X(p, q) > 1$, and $d_X(p, q) < 1$.

Case 1: $d_X(p, q) = 1$. Anything real to the power of 1 is just 1, so if we set $C = M$, then it will always be true that $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$.

Case 2: $d_X(p, q) < 1$. If $d_X(p, q) < 1$, then for $0 < \alpha \leq 1$, $d_X(p, q) \leq d_X(p, q)^\alpha$. Therefore, if we set $C = L$, then $L d_X(p, q) \leq L d_X(p, q)^\alpha$, and therefore $d_Y(f(p), f(q)) \leq C d_X(p, q)^\alpha$.

Case 3: $d_X(p, q) > 1$. If $d_X(p, q) > 1$, then for all $0 < \alpha \leq 1$, $d_X(p, q)^\alpha > 1$. Therefore, if we set $C = M$, then $C d_X(p, q)^\alpha > M$, and by extension it will always be true that $d_Y(f(p), f(q)) < C d_X(p, q)^\alpha$.

Therefore, depending on the value of our inserts, we can take $C = \max\{M, L\}$ as a constant and f will be Hölder continuous. OEΔ

1.4 Solution (d)

Proof. $[a, b]$ is a subset of the real numbers, so we will be using the usual metric. If f is C^1 , then its first derivative exists and is continuous on $[a, b]$. Intervals are compact in the real numbers, so $[a, b]$ is compact. Since continuous functions on compact sets are bounded, $f'([a, b])$ is bounded. There exists $M \in \mathbb{R}$ such that for all $x \in [a, b]$, $|f'(x)| < M$. Without loss of generality, suppose $p \leq q$. By the mean value theorem, there exists $c \in (p, q)$ so that $f'(c) = \frac{f(q) - f(p)}{q - p}$. Since $c \in [a, b]$, $|f'(x)| < M$, so $\left| \frac{f(q) - f(p)}{q - p} \right| < M$. By the limit laws, $\frac{|f(q) - f(p)|}{|q - p|} < M$. Multiply both sides by $|q - p|$ to find that $|f(q) - f(p)| < M |q - p|$. Since $\mathbb{R}$ uses the usual metric, this means for any $x, p \in X$, $d_{\mathbb{R}}(f(p), f(q)) \leq L d_{[a, b]}(p, q)$ for any $p, q \in [a, b]$. Therefore, f is Lipschitz continuous. OEΔ

1.5 Solution (e)

Proof. Let f be Hölder continuous with exponent $\alpha > 1$. There exists a constant C such that for all $p, q \in \mathbb{R}$:

$$d_{\mathbb{R}}(f(p), f(q)) \leq C d_{\mathbb{R}}(p, q)^\alpha \quad (21)$$

Note that $\mathbb{R}$ uses the usual metric.

$$|f(p) - f(q)| \leq C |p - q|^\alpha \quad (22)$$

Divide both sides by $|p - q|$.

$$\frac{|f(p) - f(q)|}{|p - q|} = \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (23)$$

By the definition of the absolute value, $\left| \frac{f(p) - f(q)}{p - q} \right| \geq 0$.

$$0 \leq \left| \frac{f(p) - f(q)}{p - q} \right| \leq C |p - q|^{\alpha-1} \quad (24)$$

Take the limit of all sides as q approaches p .

$$\lim_{q \rightarrow p} 0 \leq \lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| \leq \lim_{q \rightarrow p} C |p - q|^{\alpha-1} \quad (25)$$

We proved in *0.1: Limit Proof I* that $\lim_{q \rightarrow p} C |p - q|^{\alpha-1} = 0$. Since 0 is constant and contains neither p nor q , $\lim_{q \rightarrow p} 0 = 0$. Therefore, by the squeeze theorem, $\lim_{q \rightarrow p} \left| \frac{f(p) - f(q)}{p - q} \right| = f'(p) = 0$ for all $p \in \mathbb{R}$.

Now, recall the mean value theorem: if f is a real, continuous function on $[a, b]$ which is differentiable on (a, b) , there is a point $x \in (a, b)$ where $f(b) - f(a) = (b - a)f'(x)$. Since our $f'(x) = 0$ for all $x \in \mathbb{R}$, fix p and let $q \in \mathbb{R}$. The Mean Value Theorem tells us that for the interval $[p, q]$, there exists a point $x \in \mathbb{R}$ such that $f(q) - f(p) = (q - p)f'(x) = (q - p) * 0 = 0$. Therefore, for any $q \in \mathbb{R}$, $f(p) = f(q)$. Ergo, f is constant. OEΔ

2 Problem 2

On the interval $(100, \infty)$, define functions $f(x) = x + \sin(x)$, $g(x) = x$, $p(x) = x + \sin(x) \cos(x)$, and $q(x) = p(x)e^{\sin(x)}$.

(a) Using facts we proved about limits and derivatives, prove that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1, \quad \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} \text{ DNE}, \quad \lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} \text{ DNE}, \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0.$$

(b) For both scenarios $\frac{f(x)}{g(x)}$ and $\frac{p(x)}{q(x)}$, explain why these results do not contradict L'Hôpital's rule.

Note: even though we haven't learned about sin and cos yet, you are free to use any familiar properties of them you wish.

2.1 Solution (a)

We prove that $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$.

Proof. First write out the limit.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{x + \sin(x)}{x} = \lim_{x \rightarrow \infty} \left[1 + \frac{\sin(x)}{x} \right] \quad (26)$$

By limit laws, we can separated added up properties.

$$\lim_{x \rightarrow \infty} \left[1 + \frac{\sin(x)}{x} \right] = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \quad (27)$$

Since 1 is constant, $\lim_{x \rightarrow \infty} 1 = 1$. My first assertion about sin: it is bounded to $[-1, 1]$ and is cyclical, so $-1 \leq \sin(x) \leq 1$. Divide all sides by x to find $-\frac{1}{x} \leq \frac{\sin(x)}{x} \leq \frac{1}{x}$. Take the limit as x approaches ∞ to get $-\lim_{x \rightarrow \infty} \frac{1}{x} = \lim_{x \rightarrow \infty} -\frac{1}{x} \leq \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} \leq \lim_{x \rightarrow \infty} \frac{1}{x}$. We proved earlier that $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$, and by extension, $-\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Therefore, by the sequence squeeze theorem, $\lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 0$. Put this back into $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$.

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{\sin(x)}{x} = 1 + 0 = 1 \quad (28)$$

OEΔ

We prove that $\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$ does not exist.

Proof. My second assertion is that if $f(x) = \sin(x)$, $f'(x) = \cos(x)$. Since $(a + b)' = a' + b'$, $(x + \sin(x))' = (x') + (\sin(x))' = 1 + \cos(x)$.

$$\lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \infty} [1 + \cos(x)] \quad (29)$$

Create two sequences in terms of $n \in \mathbb{N}$ that diverge to ∞ as $n \rightarrow \infty$.

$$a_n = 2n\pi; b_n = (2n + 1)\pi \quad (30)$$

Apply both a_n and b_n to $1 + \cos(x)$.

$$1 + \cos(a_n) = 1 + \cos(2n\pi) = 1 + 1 = 2 \quad (31)$$

$$1 + \cos(b_n) = 1 + \cos((2n+1)\pi) = 1 - 1 = 0 \quad (32)$$

This gives us two subsequences that converge to two different values as $x \rightarrow \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)}$ does not exist.

Proof.

$$\lim_{x \rightarrow \infty} \frac{p(x)}{q(x)} = \lim_{x \rightarrow \infty} \frac{p(x)}{p(x)e^{\sin(x)}} = \lim_{x \rightarrow \infty} e^{-\sin(x)} \quad (33)$$

Define two sequences as $n \rightarrow \infty$.

$$a_n = \left(2n + \frac{1}{2}\right)\pi; b_n = \left(2n + \frac{3}{2}\right)\pi$$

Apply both a_n and b_n to $e^{-\sin(x)}$.

$$e^{-\sin(a_n)} = e^{-\sin((2n+\frac{1}{2})\pi)} = e^{-1} \quad (34)$$

$$e^{-\sin(b_n)} = e^{-\sin((2n+\frac{3}{2})\pi)} = e^1 = e \quad (35)$$

This gives us two subsequences that converge to two different values as $x \rightarrow \infty$. Therefore, the limit does not exist. OEΔ

We prove that $\lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0$.

Proof. First, we find the derivative of $p(x)$ and $q(x)$. This involves the multiplication rule $((fg)' = fg' + f'g)$, the chain rule $((f(g))' = g'f'(g))$, my assertion that $(\cos(x))' = -\sin(x)$, and the $\frac{d}{dx}e^x = e^x$. I was given verbal approval in Office Hours to assert that $\frac{d}{dx}e^x = e^x$.

$$p'(x) = (x + \sin(x) \cos(x))' = 1 + (\sin(x) \cos(x))' = 1 + \cos^2(x) - \sin^2(x) \quad (36)$$

$$= 1 + \cos(2x) = 2 \cos^2(x) \quad (37)$$

$$q'(x) = \left(p(x)e^{\sin(x)}\right)' = p'(x)e^{\sin(x)} + p(x) \left(e^{\sin(x)}\right)' \quad (38)$$

$$= p'(x)e^{\sin(x)} + p(x) \cos(x)e^{\sin(x)} \quad (39)$$

$$= 2 \cos^2(x)e^{\sin(x)} + (x + \sin(x) \cos(x)) \cos(x)e^{\sin(x)} \quad (40)$$

$$= e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x)) \quad (41)$$

Now, we compute $\frac{p'(x)}{q'(x)}$.

$$\frac{p'(x)}{q'(x)} = \frac{2 \cos^2(x)}{e^{\sin(x)} \cos(x) (2 \cos(x) + x + \sin(x) \cos(x))} \quad (42)$$

$$= \frac{2 \cos(x)}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \quad (43)$$

Take a few inequalities based on the bounds of $\sin(x)$ and $\cos(x)$ being $-1 \leq \sin(x) \leq 1$ and $-1 \leq \cos(x) \leq 1$.

$$e^{-1} \leq e^{\sin(x)} \leq e \quad (44)$$

$$-2 \leq 2 \cos(x) \leq 2 \quad (45)$$

$$-1 \leq \sin(x) \cos(x) \leq 1 \quad (46)$$

$$-3 \leq 2 \cos(x) + \sin(x) \cos(x) \leq 3 \quad (47)$$

$$x - 3 \leq 2 \cos(x) + x + \sin(x) \cos(x) \leq x + 3 \quad (48)$$

$$e^{-1} (x - 3) \leq e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x)) \leq e (x + 3) \quad (49)$$

$$\frac{1}{e^{-1} (x - 3)} \geq \frac{1}{e^{\sin(x)} (2 \cos(x) + x + \sin(x) \cos(x))} \geq \frac{1}{e (x + 3)} \quad (50)$$

It can be proven that $\frac{1}{e^{-1}(x-3)}$ converges to 0 and $\frac{1}{e(x+3)}$ converges to 0, which would mean that $\frac{1}{e^{\sin(x)}(2 \cos(x) + x + \sin(x) \cos(x))}$ converges to 0. $-1 \leq \cos(x) \leq 1$, meaning $-2 \leq 2 \cos(x) \leq 2$, meaning $-\frac{2}{e^{\sin(x)}(2 \cos(x) + x + \sin(x) \cos(x))} \leq \frac{2 \cos(x)}{e^{\sin(x)}(2 \cos(x) + x + \sin(x) \cos(x))} \leq \frac{2}{e^{\sin(x)}(2 \cos(x) + x + \sin(x) \cos(x))}$. Both of the side values converge to 0 since multiplying by a constant does not change convergence to 0. Therefore, $\frac{2 \cos(x)}{e^{\sin(x)}(2 \cos(x) + x + \sin(x) \cos(x))}$ converges to 0. Ergo, $\lim_{x \rightarrow \infty} \frac{p'(x)}{q'(x)} = 0$. OEΔ

2.2 Solution (b)

In the case of $\frac{f(x)}{g(x)}$, we do know the limit. However, l'Hôpital's rule does not guarantee that the limit of the ratio of the derivatives exists if the limit of the ratio exists and fulfills the requirements, just that the limit of the ratio exists if the limit of the ratio of the derivatives exists and fulfills the requirements. In short, the converse of l'Hôpital's rule does not apply.

In the case of $\frac{p(x)}{q(x)}$, the continuous cyclical nature of $\cos(x)$ between -1 and 1 , and since $q'(x)$ has $\cos(x)$ as a multiplicand, $q'(x)$ is equal to 0 at infinite points on the interval $(100, \infty)$. This goes against the requirement for l'Hôpital's rule that the denominator is not equal to 0 at any point in the interval considered.

3 Problem 3

Let $f : (a, b) \rightarrow \mathbb{R}$ be continuous. Let $c \in (a, b)$, and suppose f is differentiable on (a, c) and (c, b) . Suppose that

$$\lim_{x \rightarrow c} f'(x)$$

exists and is a real number. Show that f is differentiable at c . *Hint:* Mean value theorem.

3.1 Solution

Proof. Let $\lim_{x \rightarrow c} f'(x) = L$. Recall the definition of $f'(c)$.

$$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c} \quad (51)$$

Without loss of generality, suppose that $t > c$. By the Mean Value Theorem, there exists $d \in (c, t)$ where $\frac{f(t) - f(c)}{t - c} = f'(d)$. Since $d \in (c, t)$, $c < d < t$, so $0 < d - c < t - c$ and therefore $0 < |d - c| < |t - c|$.

$f'(c) = \lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c}$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ where $|t - c| < \delta$ implies $\left| \frac{f(t) - f(c)}{t - c} - f'(c) \right| < \varepsilon$. Since $|d - c| < |t - c|$, $|d - c| < \delta$. Furthermore, since $\frac{f(t) - f(c)}{t - c} = f'(d)$, $|f'(d) - f'(c)| < \varepsilon$. Therefore, for all $\varepsilon > 0$, there exists $\delta > 0$ where $|d - c| < \delta$ implies $|f'(d) - f'(c)| < \varepsilon$. This means $f'(c) = \lim_{d \rightarrow c} f'(d)$. It is given that $\lim_{x \rightarrow c} f'(x)$ exists and is a real number. Therefore $f'(c)$ exists and is a real number. Ergo, f is differentiable at c . $\square$

4 Problem 4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^2 (meaning its second derivative exists and is continuous on $\mathbb{R}$).

- (a) Let $x \in \mathbb{R}$. For each $h > 0$, show there are numbers ξ and η with $x - h < \xi < x < \eta < x + h$ and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (b) Show that, for each $x \in \mathbb{R}$,

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

- (c) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *convex* if for all $x, y \in \mathbb{R}$ and $0 \leq \lambda \leq 1$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Geometrically, this means the graph of f lies *below* all of its secant lines. Show that any C^2 convex function f satisfies $f''(x) \geq 0$ for all x . *Hint:* convexity tells you something about the numerator in part (b).

- (d) Conversely, if a twice-differentiable function f satisfies $f''(x) \geq 0$ for all x , show that f is convex. *Hint:* For $\lambda \in (0, 1)$ and $x < y \in \mathbb{R}$, the number $z := \lambda x + (1 - \lambda)y$ satisfies $x < z < y$. Apply the mean value theorem on $[x, z]$ and $[z, y]$ to get an expression for $\lambda f(x) + (1 - \lambda)f(y)$.

4.1 Solution (a)

Proof. Recall Taylor's Theorem, for some $\gamma \in (\alpha, \beta)$ or $\gamma \in (\beta, \alpha)$.

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k \quad (52)$$

$$f(\beta) = P(\beta) + \frac{f^{(n)}(\gamma)}{n!} (\beta - \alpha)^n \quad (53)$$

Let $n = 2$ and let $\alpha = x$ and $\beta = x + h$. Some $\eta \in (x, x + h)$ will satisfy Taylor's Theorem for this.

$$P(t) = f(x) + f'(x)(t - x) \quad (54)$$

$$f(x + h) = f(x) + f'(x)h + \frac{f''(\eta)h^2}{2} \quad (55)$$

Let $n = 2$ and let $\alpha = x$ and $\beta = x - h$. Some $\xi \in (x - h, x)$ will satisfy Taylor's Theorem for this.

$$P(t) = f(x) + f'(x)(t - x) \quad (56)$$

$$f(x - h) = f(x) - f'(x)h + \frac{f''(\xi)h^2}{2} \quad (57)$$

Add the latter two of each of these equalities together.

$$f(x + h) + f(x - h) = f(x) + f'(x)h + \frac{f''(\eta)h^2}{2} + f(x) - f'(x)h + \frac{f''(\xi)h^2}{2} \quad (58)$$

Rearrange this to isolate $\frac{(f''(\eta)+f''(\xi))h^2}{2}$, then divide both sides by h^2 .

$$f(x+h) + f(x-h) - f(x) - f(x) - f'(x)h + f'(x)h = \frac{f''(\eta)h^2}{2} + \frac{f''(\xi)h^2}{2} \quad (59)$$

$$f(x+h) + f(x-h) - 2f(x) = \frac{(f''(\eta) + f''(\xi))h^2}{2} \quad (60)$$

$$\frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = \frac{f''(\eta) + f''(\xi)}{2} \quad (61)$$

This is the equation we were looking for.

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4.2 Solution (b)

Proof. We know that for each $h > 0$, show there are numbers ξ and η with $x-h < \xi < x < \eta < x+h$ and

$$\frac{f''(\xi) + f''(\eta)}{2} = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}. \quad (62)$$

$f : \mathbb{R} \rightarrow \mathbb{R}$ and $f'' : \mathbb{R} \rightarrow \mathbb{R}$ are both continuous. This means that for all $\varepsilon > 0$, there exists $\delta > 0$ where $0 < |x+h-x| = |h| < \delta$ (or $0 < |x-h-x| = |h| < \delta$) implies $|f''(x+h) - f''(x)| < \varepsilon$ (or $|f''(x-h) - f''(x)| < \varepsilon$). Since $x-h < \xi < x$, $x-h-x < \xi-x < 0$ and therefore $|\xi-x| < |h|$ and therefore $|f''(\xi) - f''(x)| < \varepsilon$. Similarly, we can say that for the η prior where $x < \eta < x+h$, $|f''(\eta) - f''(x)| < \varepsilon$. Add these together to find that $|f''(\xi) - f''(x)| + |f''(\eta) - f''(x)| < 2\varepsilon$. By the triangle inequality, $|f''(\xi) + f''(\eta) - 2f''(x)| < 2\varepsilon$. Divide both sides by 2 and find $\left| \frac{f''(\xi) + f''(\eta)}{2} - f''(x) \right| < \varepsilon$. Using the same value of h as before and using (62), we find that $|h| < \delta$ implies that $\left| \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} - f''(x) \right| < \varepsilon$.

$$\text{Ergo, } \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x).$$

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4.3 Solution (c)

Proof. Pick $m, h \in \mathbb{R}$ where $h > 0$ and let $x = m+h$ and $y = m-h$.

$$f(\lambda(m+h) + (1-\lambda)(m-h)) \leq \lambda f(m+h) + (1-\lambda)f(m-h) \quad (63)$$

Since this holds for all $\lambda \in [0, 1]$, choose $\lambda = \frac{1}{2}$, then multiply all sides by 2.

$$f\left(\frac{m+h}{2} + \frac{m-h}{2}\right) \leq \frac{f(m+h)}{2} + \frac{f(m-h)}{2} \quad (64)$$

$$0 \leq f(m+h) + f(m-h) - 2f(m) \quad (65)$$

Since $h > 0$, we can divide both sides by h^2 .

$$0 \leq \frac{f(m+h) + f(m-h) - 2f(m)}{h^2} \quad (66)$$

This applies for all $h > 0$. Using what we proved in part (b), we can say that $f''(m) = \lim_{h \rightarrow 0} \frac{f(m+h) + f(m-h) - 2f(m)}{h^2}$. Therefore, $0 \leq f''(m)$. OEΔ

4.4 Solution (d)

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f''(x) \geq 0$ for all $x \in \mathbb{R}$. To get it out of the way, if $\lambda = 0$ or $\lambda = 1$, the convexity equation would become $f(x) \leq f(x)$ or $f(y) \leq f(y)$ respectively, which are always true. Therefore, let $0 < \lambda < 1$ and let $x < y$. Let $z = \lambda x + (1 - \lambda)y$. By 0.6: *Lambda in the Middle*, $x < z < y$.

Let's apply Taylor's Theorem here, first where $\alpha = z$ and $\beta = x$ for $n = 2$.

$$P(t) = f(z) + f'(z)(t - z) \quad (67)$$

$$f(x) = f(z) + f'(z)(x - z) + \frac{f''(\eta)(x - z)^2}{2} \quad (68)$$

$$f(x) = f(\lambda x + (1 - \lambda)y) + f'(\lambda x + (1 - \lambda)y)((1 - \lambda)x + (\lambda - 1)y) + \frac{f''(\eta)((\lambda - 1)x + (1 - \lambda)y)^2}{2} \quad (69)$$

Apply the inequality that $f''(\eta) \geq 0$.

$$f(x) \geq f(z) + f'(z)(x - z) \quad (70)$$

Now do so where $\alpha = z$ and $\beta = y$ for $n = 2$.

$$f(y) = f(z) + f'(z)(y - z) + \frac{f''(\xi)(y - z)^2}{2} \quad (71)$$

$$f(y) = f(z) + f'(z)(\lambda y - \lambda x) + \frac{f''(\xi)(\lambda y - \lambda x)^2}{2} \quad (72)$$

Apply the inequality that $f''(\xi) \geq 0$.

$$f(y) \geq f(z) + f'(z)(y - z) \quad (73)$$

Multiply (70) by λ and (73) by $1 - \lambda$.

$$\lambda f(x) \geq \lambda f(z) + \lambda f'(z)(x - z) \quad (74)$$

$$(1 - \lambda)f(y) \geq (1 - \lambda)f(z) + (1 - \lambda)f'(z)(y - z) \quad (75)$$

Now add (74) and (75) together. From there, we can cancel things out.

$$\lambda f(x) + (1 - \lambda)f(y) \geq \lambda f(z) + (1 - \lambda)f(z) + \lambda f'(z)(x - z) + f'(z)(y - z) - \lambda f'(z)(y - z) \quad (76)$$

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + \lambda f'(z)(x - y) + f'(z)(y - z) \quad (77)$$

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + \lambda f'(z)(x - y) + f'(z)(y - (\lambda x + (1 - \lambda)y)) \quad (78)$$

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + \lambda f'(z)(x - y) + f'(z)(\lambda y - \lambda x) \quad (79)$$

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) \quad (80)$$

Ergo, f is convex.

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5 Problem 5

Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Compute the upper and lower Riemann integrals of f . Is f Riemann integrable?

5.1 Solution

The issue here is the density of the rationals in the reals.

First we prove the upper Riemann integral.

$$\overline{\int_0^1} f(x) dx = 1 \quad (81)$$

Proof. Let P be any partition of $[0, 1]$ with $n + 1$ elements. Take any $j \in \mathbb{N}$ where $1 \leq j \leq n$. By the density of the rationals in the reals, there exists $x \in [x_{j-1}, x_j]$ where $x \in \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\sup_{x \in [x_{j-1}, x_j]} f(x) = 1$. We can compute the upper sum of f from here.

$$U(P, f) = \sum_{j=1}^n \Delta x_j \sup_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 1 = \sum_{j=1}^n \Delta x_j \quad (82)$$

Since the sum of Δx_j telescopes, $\sum_{j=1}^n \Delta x_j = x_n - x_0$. Therefore, for any partition P of $[0, 1]$, $U(P, f) = 1 - 0 = 1$. Since $U(P, f) = 1$ for every partition P of $[0, 1]$, $\inf\{U(P, f) | \text{Partition } P\} = 1$. Ergo, $\overline{\int_0^1} f(x) dx = \inf\{U(P, f) | \text{Partition } P\} = 1$. OEΔ

Now we prove the lower Riemann integral.

$$\underline{\int_0^1} f(x) dx = 0 \quad (83)$$

Proof. Let P be any partition of $[0, 1]$ with $n + 1$ elements. Take any $j \in \mathbb{N}$ where $1 \leq j \leq n$. By the density of the irrationals in the reals, there exists $t \in [x_{j-1}, x_j]$ where $t \notin \mathbb{Q}$. Since the only two possible values of $f(x)$ are 0 and 1, $\inf_{t \in [x_{j-1}, x_j]} f(t) = 0$. We can compute the upper sum of f from here.

$$L(P, f) = \sum_{j=1}^n \Delta x_j \inf_{t \in [x_{j-1}, x_j]} f(t) = \sum_{j=1}^n \Delta x_j * 0 = \sum_{j=1}^n 0 = 0 \quad (84)$$

Therefore, for any partition P of $[0, 1]$, $L(P, f) = 0$. Since $L(P, f) = 0$ for every partition P of $[0, 1]$, $\sup\{L(P, f) | \text{Partition } P\} = 0$. Ergo, $\underline{\int_0^1} f(x) dx = \sup\{L(P, f) | \text{Partition } P\} = 0$. OEΔ

Since $\overline{\int_0^1} f(x) dx \neq \underline{\int_0^1} f(x) dx$, the function is not Riemann integrable on $[0, 1]$.